

Hanger Counter Derivation of Equations

Objective: Find the number of resistors in parallel, n , in a circuit with one resistor in series, R_0 .

Assume: All resistors in parallel have the same resistance as R_0 .

- ① Find the total resistance from the voltage drop across R_0 ,

$$V_{R_0} = R_0 i$$

$$i = \frac{V_T}{R_T}$$

$$\Rightarrow V_{R_0} = R_0 \left(\frac{V_T}{R_T} \right)$$

$$\therefore R_T = R_0 \left(\frac{V_T}{V_{R_0}} \right) \quad [1]$$

- ② Find the number of resistors in parallel, n , using the total resistance, R_T ,

$$R_T = R_0 + \left(\sum_{j=1}^n \frac{1}{R_j} \right)^{-1}$$

$$\text{Apply } R_0 = R_1 = R_2 = \dots = R_n$$

$$\Rightarrow R_T = R_0 + \left(\frac{n}{R_0} \right)^{-1}$$

$$\Rightarrow R_T = R_0 + \left(\frac{R_0}{n} \right)$$

$$\therefore n = \frac{R_0}{R_T - R_0} \quad [2]$$

- ③ For an Arduino with a 5V output and $R_0 = 330 \Omega$, we have:

$$[1] \quad R_T = 330 \left(\frac{5}{V_{R_0}} \right)$$

$$[2] \quad n = \frac{330}{R_T - 330}$$

* V_{R_0} is calculated as $V_T - V_{R_i}$ due to the location of the signal wire, which measures remaining voltage after R_0

* The Arduino `analogRead()` command puts voltage on a 0-1023 scale, such that $5V \rightarrow 1023$